

Sampling Distributions

Paulo Fagandini

Lisbon Accounting and Business School – Polytechnic University of Lisbon

Sampling Distributions

Random (Casual) Sample and Statistic

Population

- Parameters (unknown)

Sample

- Statistics (known)

Statistical Inference

Statistical inference consists of a set of methods that allow us to draw conclusions about a population, based on a random (or casual) sample drawn from that population.

Random Sample, Parameters, and Statistics

A sample of size n , drawn from a population according to a random (or casual) process, is called a **random sample** if the values of the random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed. The term X_1 denotes the first element of the sample, X_2 the second, and X_n the n -th.

Parameters and Statistics

A **parameter** is a value that characterizes a given population which, although unknown, is fixed.

A **statistic** is a characteristic that is **a function of the sample values**, i.e., **it is a random variable** that does not involve any unknown parameter, denoted by $T(X_1, X_2, \dots, X_n)$.

Statistics and Estimates

Once a particular sample $x_1, x_2, \dots, x_n$ has been observed, it is possible to obtain the value of the statistic associated with this sample, namely $T(x_1, x_2, \dots, x_n)$, yielding an **estimate** for the parameter.

From each random sample drawn from a population, statistics with different values can be obtained.

Statistics are random variables, and therefore their probabilistic behavior is given by their respective distribution function (density or probability function), leading to the **sampling distributions** of the statistic represented by $T(X_1, X_2, \dots, X_n)$.

Exercise I

Consider $X_1, X_2, \dots, X_n$, $n \in \mathbb{N}$, a random sample from a $Normal(\mu = -2, \sigma)$ population, where σ is unknown. The following r.v.'s are **not** statistics:

a. [T] [F] F $T(X_1, X_2, \dots, X_n) = X_1 + X_2 + \dots + X_n$

b. [T] [F] F $T(X_1, X_2, \dots, X_n) = \bar{X}$

c. [T] [F] T $T(X_1, X_2, \dots, X_n) = \frac{\bar{X}}{\sigma}$

d. [T] [F] F $T(X_1, X_2, \dots, X_n) = \frac{\sum (X_i - \mu)^2}{n}$

e. [T] [F] F $T(X_1, X_2, \dots, X_n) = \frac{\sum (X_i - \bar{X})^2}{n - 1}$

f. [T] [F] T $T(X_1, X_2, \dots, X_n) = \sum \left(\frac{X_i - \mu}{\sigma} \right)^2$

Exercise II

Let $X_1, X_2, \dots, X_n$, $n \in \mathbb{N}$, be a random sample from a population X (with unknown probabilistic behavior), with expected value μ and variance σ^2 , both known.

- i. The expected value of the statistics $\sum_{i=1}^n X_i$ and $\bar{X}$ are, respectively, equal to:
- a) $n\mu$ and μ b) $n\mu$ and μ/n c) μ and μ/n d) none of the above

Answer: (a)

- ii. The variance of the statistics $\sum_{i=1}^n X_i$ and $\bar{X}$ are, respectively, equal to:
- a. $n\sigma^2$ and σ b) $n\sigma^2$ and σ^2/n c) σ^2 and σ^2/n d) none of the above

Answer: (b)

Chi-Square Distribution

A pertinent question arises: How to obtain the sampling distribution of a statistic or of a random variable?

Before answering this question, it is important to introduce two **fundamental theoretical distributions** within the scope of inference.

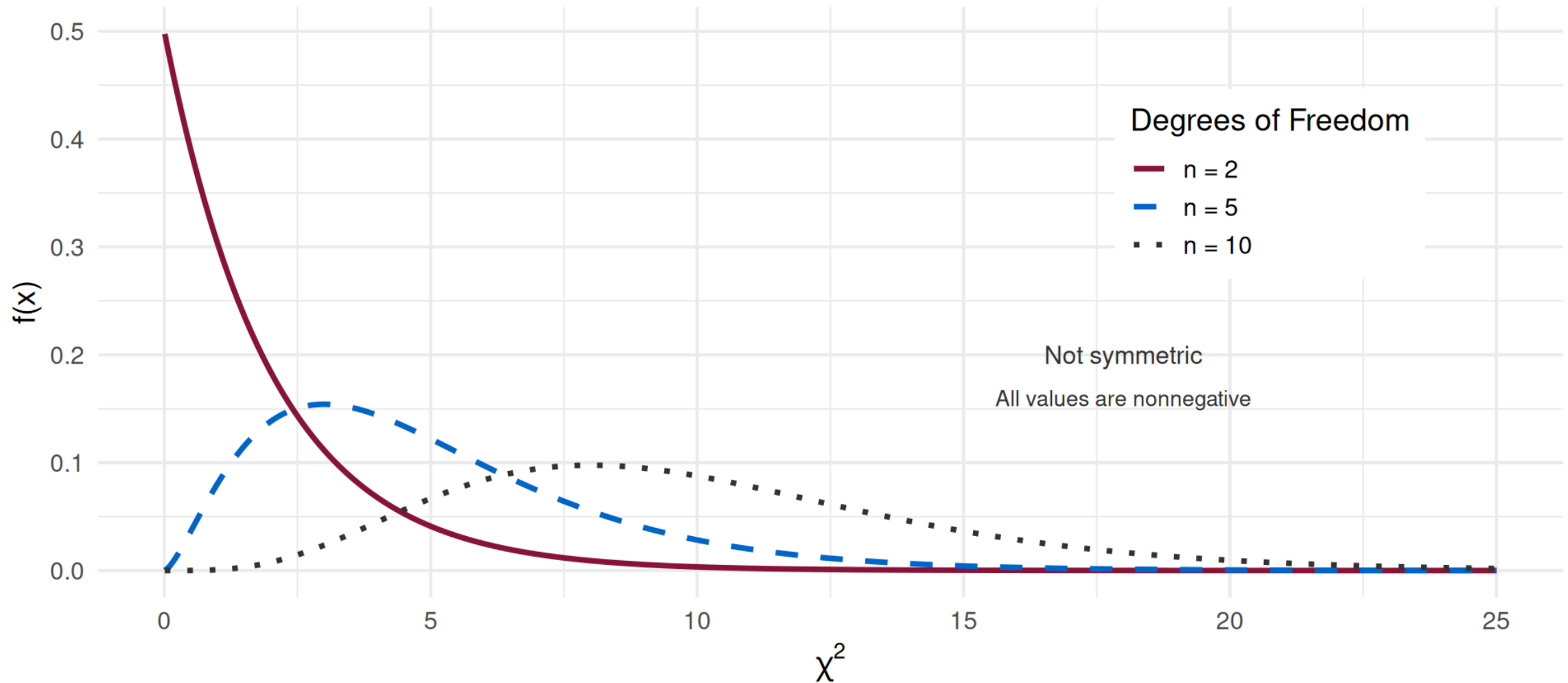
Chi-Square Distribution

Let $Z_1, Z_2, \dots, Z_n$ be independent and identically distributed random variables, $Z_i \sim \text{Normal}(0, 1)$.

$$Y = \sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2$$

is a Chi-square random variable with n degrees of freedom, i.e.: $Y \sim \chi_{(n)}^2$.

Chi-Square Distribution: Main Properties (1)



Chi-Square Distribution: Main Properties (2)

1. The graphical shape of the distribution depends on its degrees of freedom: n ($n \in \mathbb{N}$).
2. It is a positive and non-symmetric function. The skewness decreases as the degrees of freedom increase.
3. If $X \sim \chi^2_{(n)}$, then: $E(X) = n$ and $Var(X) = 2n$.
4. It is an **additive** distribution. The sum of k independent random variables, where $X_i \sim \chi^2_{(n_i)}$, is still a random variable such that:

$$\sum_{i=1}^k X_i \sim \chi^2_{\left(n=\sum_{i=1}^k n_i\right)}$$

5. With increasing degrees of freedom, the Chi-square distribution tends to the Normal distribution, by the C.L.T.:

$$\chi^2_{(n)} \stackrel{\text{CLT}}{\underset{\sim}{\leadsto}} N(n, \sqrt{2n}) \iff Z_n = \frac{\chi^2_{(n)} - n}{\sqrt{2n}} \stackrel{\text{CLT}}{\underset{\sim}{\leadsto}} N(0, 1)$$

Exercise V

Let $(Z_1, Z_2, Z_3, Z_4, Z_5)$ be a random sample from a $N(0, 1)$ population and $Q = \sum_{i=1}^5 Z_i^2$.

a. [V][F] $P(1.610 < Q < 9.236) = 0.8$

b. [V][F] The standard deviation of $Z_3^2 + Z_4^2$ is equal to 4.

a)

$$Q \sim \chi_{(5)}^2$$

$$P(1.610 < Q < 9.236) = P(Q > 1.610) - P(Q > 9.236) = 0.9 - 0.1 = 0.8$$

(direct observation from the χ^2 table)

The statement is **true**.

b)

$$Z_3^2 \sim \chi_{(1)}^2 ; Z_4^2 \sim \chi_{(1)}^2 ; Z_3^2 + Z_4^2 \sim \chi_{(2)}^2 \text{ (Additivity Theorem)}$$

$$\text{Var}(Z_3^2 + Z_4^2) = 2 \times 2 = 4 \implies \sigma(Z_3^2 + Z_4^2) = \sqrt{4} = 2$$

Statistics II

The statement is **false**

Student's t Distribution

Let $X \sim \text{Normal}(0, 1)$ and $Y \sim \chi^2_{(n)}$ be independent random variables. Then the random variable:

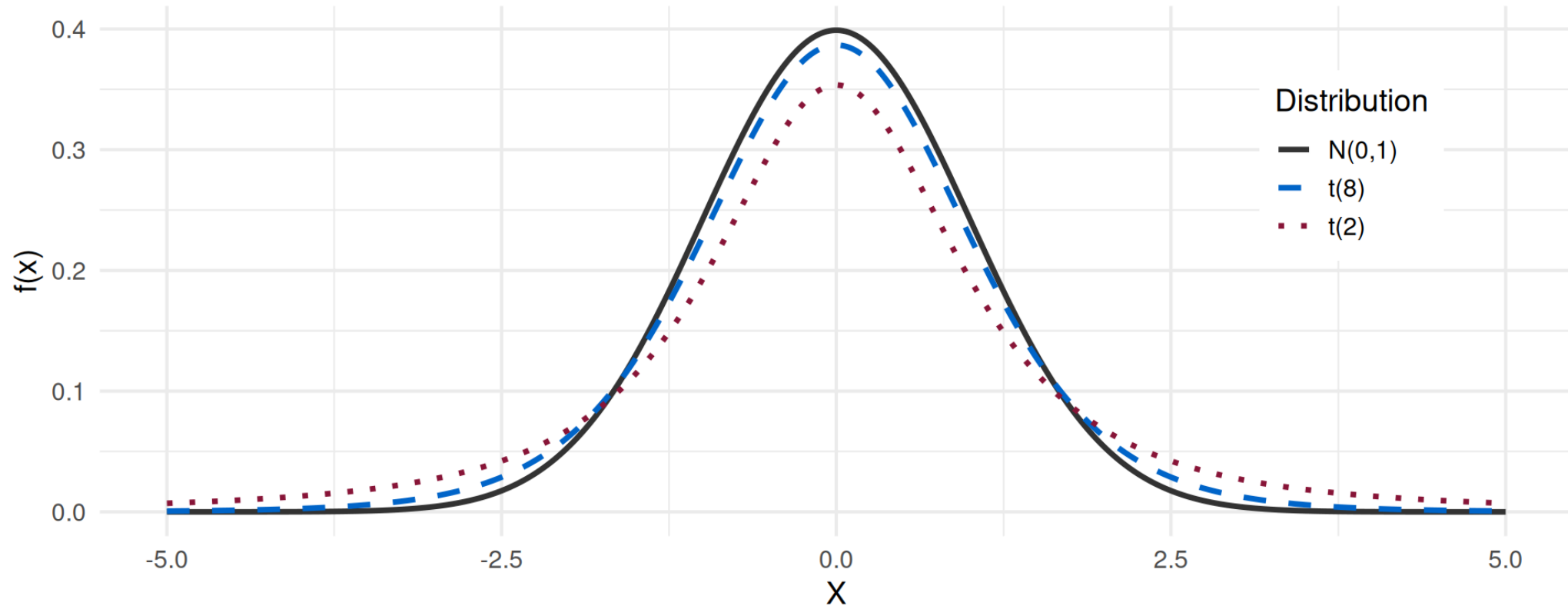
$$T = \frac{X}{\sqrt{Y/n}} \sim t_{(n)}$$

has a Student's t distribution with n degrees of freedom, i.e.: $T \sim t_{(n)}$.

Main properties of the Student's t distribution

1. The graphical shape depends on its degrees of freedom (n). As the d.f. increase, the quantiles of the Student's t converge to the corresponding quantiles of the Standard Normal.

Student's t Distribution: Properties



2. It is a **symmetric** function around the mean.

3. If $X \sim t_{(n)}$, then: $E(X) = 0$ and $Var(X) = \frac{n}{n-2}$, with $n > 2$.

Student's t Distribution: CLT Approximation

4. With increasing degrees of freedom, the Student's t distribution tends to the Normal distribution, by the C.L.T.:

$$X \overset{\text{CLT}}{\rightsquigarrow} N\left(0, \frac{n}{n-2}\right) \iff Z_n = \frac{X}{\sqrt{\frac{n}{n-2}}} \overset{\text{CLT}}{\rightsquigarrow} N(0, 1)$$

Example

Exercise 4: Let T be a r.v distributed $t - Student$ with 11 d.o.f., $T \sim t(11)$.

- i. [T][F] The value of k such that $P(T \leq k) = 0.75$ is equal to 0.303.
- ii. [T][F] The value of the probability $P(|T| > 2.201)$ is:
 - a) 0.025 b) 0.95 c) 0.05 d) None
- i. $P(T \leq k) = 0.75 \Leftrightarrow P(T > k) = 0.25 \Leftrightarrow k = 0.697$ directly from the table, and the sentence is **False**.
- ii. $P(|T| > 2.201) = 2P(T > 2.201) = 2 \times 0.025 = 0.05$ by symmetry of the distribution and direct observation of the table. The correct answer is **c**.

Sampling Distributions of the Sample Mean $\bar{X}$

Case 1: Normal Population, σ^2 Known

Population: $X \sim N(\mu, \sigma)$, with σ^2 known.

From a random sample $X_1, X_2, \dots, X_n$, where $X_i \sim N(\mu, \sigma)$, we obtain the statistic sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

The sampling distribution of $\bar{X}$ is still Normal, by the additivity theorem of the Normal distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \iff Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Case 2: Normal Population, σ^2 Unknown

Population: $X \sim N(\mu, \sigma)$, with σ^2 unknown.

Statistic: Mean of a random sample of size n : $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Sampling distribution of $\bar{X}$:

$$T = \frac{\frac{\bar{X} - \mu}{S}}{\frac{1}{\sqrt{n-1}}} = \frac{\bar{X} - \mu}{\frac{S'}{\sqrt{n}}} \sim t_{(n-1)}$$

Note: It is rare that one needs to infer the mean value when the variance is known. Thus, it is necessary to estimate the parameter σ^2 using the sample variance S^2 or the corrected sample variance S'^2 .

Case 2 (cont.) and Case 3

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{and} \quad S'^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n}{n-1} S^2$$

Approximate sampling distribution of $\bar{X}$: for n sufficiently large ($n \geq 30$), by the C.L.T.:

$$Z = \frac{\frac{\bar{X} - \mu}{S}}{\frac{1}{\sqrt{n-1}}} = \frac{\bar{X} - \mu}{\frac{S'}{\sqrt{n}}} \stackrel{\text{CLT}}{\sim} N(0, 1)$$

Case 3: Unknown population, with mean μ and **known** variance σ^2 .

Statistic: Mean of a random sample of size n ($n \geq 30$): $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Case 3 (cont.)

Approximate sampling distribution of $\bar{X}$: by the C.L.T.,

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \stackrel{\text{CLT}}{\sim} N(0, 1)$$

Note: If σ^2 is unknown, we have the approximate sampling distribution:

$$Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n-1}}} = \frac{\bar{X} - \mu}{\frac{S'}{\sqrt{n}}} \stackrel{\text{CLT}}{\sim} N(0, 1)$$

Exercise VII

A certain company intends to validate the value (in euros) of its accounts receivable from clients, with an underlying distributional behavior of the account values being $Normal(\mu = 385, \sigma = ?)$. A random sample of 25 client accounts (independent) was selected, yielding a corrected sample standard deviation of $s' = 122.6$ euros.

- a. V F The sampling distribution of the mean value of client accounts is $t_{(25)}$.
- b. V F The sample size to be collected (assuming the corrected sample standard deviation remains the same) so that the sample mean does not deviate from the population mean by more than 20 euros in 90% of cases is $n = 102$. (Assume, for the calculations, that the value of n to be found will be greater than 30.)

Exercise VII — Solution: a)

Population: $X \sim Normal(\mu = 385, \sigma = ?)$ (standard deviation unknown)

Sample: $n = 25 \rightarrow s' = 122.6$

a)

The sampling distribution of the statistic sample mean (mean value of client accounts) is given by:

$$T = \frac{\bar{X} - \mu}{\frac{s'}{\sqrt{n}}} \sim t_{(n-1)} \equiv \frac{\bar{X} - \mu}{\frac{s'}{\sqrt{n}}} \sim t_{(24)}$$

The statement is **false**.

Exercise VII — Solution: b)

b)

Population: $X \sim \text{Normal}(\mu = 385, \sigma = ?)$ (standard deviation unknown)

Sample: $n \geq 30 \rightarrow s' = 122.6$

$$\bar{X} \stackrel{\text{CLT}}{\sim} N\left(\mu, \frac{s'}{\sqrt{n}}\right) \equiv N\left(385, \frac{122.6}{\sqrt{n}}\right)$$

$$P(|\bar{X} - \mu| < 20) = 0.9 \iff P\left(|Z| < \frac{20}{\frac{122.6}{\sqrt{n}}}\right) = 0.9$$

$$\iff P\left(|Z| > \frac{20}{\frac{122.6}{\sqrt{n}}}\right) = 0.1$$

$$\iff \frac{20}{122.6} \stackrel{\text{Statistic II}}{=} 1.645 \iff n \geq 102$$

Sampling Distributions of the Sample Variance S^2

Sampling Distribution of S^2

Population: $X \sim N(\mu, \sigma)$, with mean μ unknown and variance σ^2 .

Selected a random sample $X_1, X_2, \dots, X_n$, where $X_i \sim N(\mu, \sigma)$, we obtain the statistic sample variance S^2 or corrected sample variance S'^2 :

Sampling distribution of S^2 or S'^2 :

$$Q = \frac{(n-1)S'^2}{\sigma^2} = \frac{nS^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Exercise VIII

A pharmaceutical company produces pills in which the variability of the quantity of the active substance from one pill to another should be very small. The standard deviation of the population is, supposedly, one milligram. Inspectors from the Ministry of Health selected a random sample of 16 pills. Assuming the population is Normal, the probability that the corrected sample variance is greater than 0.736 mg^2 is equal to:

- a. 0.76 b) 0.25 c) 0.75 d) none of the above

Population: $X \sim N(\mu, \sigma)$

Pivotal variable:

$$Q = \frac{(n-1)S'^2}{\sigma^2} = \frac{(n-1)S'^2}{1} = (n-1)S'^2 \sim \chi^2_{(n-1)} \equiv \chi^2_{(15)}$$

We want:

$$P(S'^2 > 0.736) = P(Q > 15 \times 0.736) = P(Q > 11.04) \approx 0.75$$

(direct observation from the χ^2 distribution table)

The correct answer is **c**).

Sampling Distributions of the Proportion from a Bernoulli Population

Case 1: Total of Successes

Selected a random sample $X_1, X_2, \dots, X_n$ from a Bernoulli population, where $X_i \sim \text{Bernoulli}(1, p)$, and let Y be the number of elements that possess the characteristic of interest (successes) in the sample of size n .

Case 1:

The statistic $Y = \sum_{i=1}^n X_i$ (total or absolute frequency of successes) has the sampling distribution:

$$Y = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p)$$

If n is sufficiently large, the C.L.T. justifies the approximate sampling distribution:

$$Y \overset{\text{CLT}}{\underset{\sim}{\rightsquigarrow}} N\left(np, \sqrt{np(1-p)}\right) \iff Z_n = \frac{Y - np}{\sqrt{np(1-p)}} \overset{\text{CLT}}{\underset{\sim}{\rightsquigarrow}} N(0, 1)$$

Case 2: Sample Proportion

Case 2:

Let the statistic $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{Y}{n}$, the proportion (relative frequency) of successes that, in a sample of size n , possess a given attribute.

It is important to know the sampling distribution of this statistic $\bar{X} = \hat{p}$.

$$E(\bar{X}) = E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{1}{n}E(X_1 + X_2 + \cdots + X_n) = \frac{1}{n}np = p$$

$$Var(\bar{X}) = Var\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{1}{n^2}Var(X_1 + \cdots + X_n) = \frac{1}{n^2}npq = \frac{p(1-p)}{n}$$

Case 2: Approximate Distribution of $\hat{p}$

When $n \rightarrow \infty$, the C.L.T. yields the approximate sampling distribution:

$$\bar{X} = \hat{p} \stackrel{\text{CLT}}{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right) \iff Z_n = \frac{\bar{X} - p}{\sqrt{\frac{p(1-p)}{n}}} \stackrel{\text{CLT}}{\sim} N(0, 1)$$

Exercise X (Statement)

In a certain neighborhood, the proportion of inhabitants who consider the opening of a new supermarket useful is 0.8. The supermarket chain *Gota Doce* intends to open a new store in this neighborhood. Nevertheless, before making a decision, a study was conducted in which 400 inhabitants were randomly interviewed to assess their intention to use the services of the new supermarket.

- a. V F The **approximate** probability that, in the sample of interviewed inhabitants, at least 300 consider the opening of the *Gota Doce* supermarket useful is equal to 0.9948.
- b. V F The probability that the deviation between the sample proportion of inhabitants who stated they would actually use the services of the new supermarket, and the true proportion, is less than 0.02, is 0.8413.

Exercise X — Solution: a)

$p = 0.8$ (proportion of inhabitants who consider the opening of a new supermarket useful)

a) $X \rightarrow$ number of inhabitants, out of 400, who consider the opening of the new supermarket useful.

$$X \sim \text{Binomial}(n = 400, p = 0.8)$$

$$\begin{aligned} P(X \geq 300) &= P(X \geq 299.5) \approx P\left(Z \geq \frac{299.5 - 400 \times 0.8}{\sqrt{400 \times 0.8 \times 0.2}}\right) \\ &= P(Z \geq -2.56) = P(Z \leq 2.56) = \Phi(2.56) = 0.9948 \end{aligned}$$

Note: 1) Continuity correction 2) CLT, approximation of the Binomial to the Normal distribution

$$Z = \frac{X - np}{\sqrt{np(1 - p)}} \dot{\sim} N(0, 1)$$

The statement is **true**.

Exercise X — Solution: b)

b) Of the interviews conducted, X inhabitants out of n expressed the opening of *Gota Doce* was useful, so the sample proportion is $\frac{X}{n}$.

$$\begin{aligned} P\left(\left|\frac{X}{n} - p\right| < 0.02\right) &\approx P\left(|Z| < \frac{0.02}{\sqrt{\frac{0.8 \times 0.2}{400}}}\right) = P(|Z| < 1) \\ &= P(-1 < Z < 1) = \Phi(1) - \Phi(-1) = \Phi(1) - [1 - \Phi(1)] \\ &= 2\Phi(1) - 1 = 2 \times 0.8413 - 1 = 0.6826 \end{aligned}$$

Note: Observe that $\frac{X - np}{\sqrt{np(1-p)}} = \frac{\frac{X}{n} - p}{\sqrt{\frac{p(1-p)}{n}}}$ and that, by the CLT,

$$Z = \frac{X - np}{\sqrt{np(1-p)}} \stackrel{\sim}{\sim} N(0, 1) \quad \text{and, therefore, also} \quad Z = \frac{\frac{X}{n} - p}{\sqrt{\frac{p(1-p)}{n}}} \stackrel{\sim}{\sim} N(0, 1)$$

Disclaimer

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